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Deep learning

Assignment -

Dataset Pizza Sales

A Pizza sales maintains it business in multiple CSV files, including orders, orders details, pizza information and pizza categories. The objective is to import these datasets into neural network.

Step 1 :- Input Features

<u>Input</u>	<u>Symbol</u>	<u>values</u>
Quantity	I_1	1.0
Time	I_2	0.48
medium size	I_3	1.0
classic category	I_4	1.0

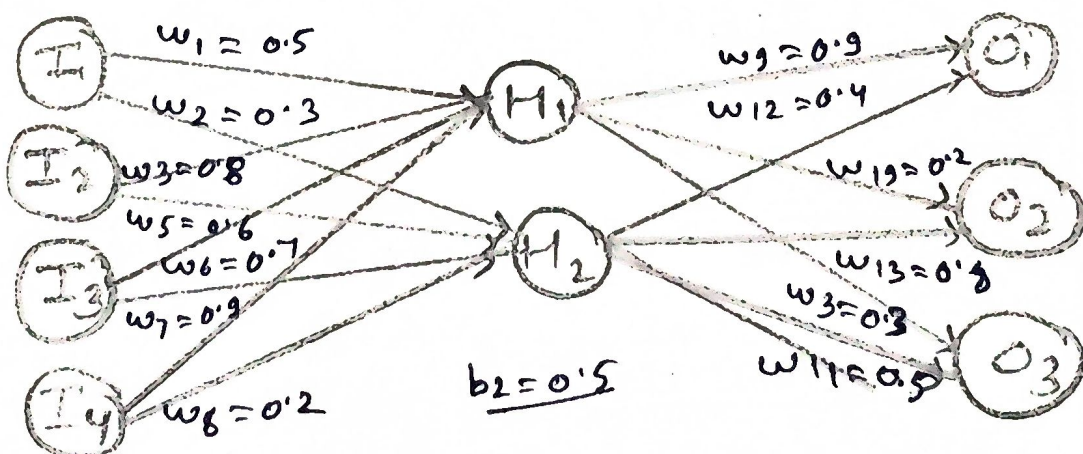
$$X = [1.0, 0.48, 1.0, 1.0]$$

Step 2 :- hidden layer weights neural networks

Input layer

hidden layer

output layer



$$b_2 = 0.1$$

activation

$$\sigma = \approx 0.947$$

$$\sigma = \approx 0.869$$

output
 .class

.ch

(4)

BOOK

① Given parameter

• Input vector(x) = [1.0, 0.48, 1.0, 1.0]

• hidden weights

From input	To H ₁ (w)	To H ₂ (w)
I ₁	0.5	0.3
I ₂	0.8	0.4
I ₃	0.6	0.7
I ₄	0.9	0.2

• Bias (hidden layer) = b₁ = 0.5

② hidden layer calculation activation

(A) Neuron H₁

$$\begin{aligned} Z_1 &= (1 \times 0.5) + (0.48 \times 0.8) + (1 \times 0.6) + (1 \times 0.9) + 0.5 \\ &= (0.5) + 0.384 + 0.6 + 0.9 + 0.5 \\ &= 2.884 \end{aligned}$$

$$\sigma = \frac{1}{1 + e^{-Z}} = \frac{1}{1 + e^{-2.884}} \approx 0.947$$

(B) Neuron H₂

$$\begin{aligned} Z_2 &= (1 \times 0.3) + (0.48 \times 0.4) + (1 \times 0.7) + (1 \times 0.2) + 0.5 \\ &= 0.3 + 0.192 + 0.7 + 0.2 + 0.5 \\ &= 1.892 \end{aligned}$$

$$\sigma = \frac{1}{1 + e^{-Z}} = \frac{1}{1 + e^{-1.892}} \approx 0.869$$

③ Output layer

$$\begin{aligned} \cdot \text{classic} = Z_c &= (0.947 \times 0.9) + (0.869 \times 0.4) + 0.1 \\ &= 1.30 \end{aligned}$$

$$\begin{aligned} \cdot \text{chicken} = Z_{ch} &= (0.947 \times 0.2) + (0.869 \times 0.8) + 0.1 \\ &= 0.98 \end{aligned}$$

$$\begin{aligned} \cdot \text{veggie} = Z_v &= (0.947 \times 0.3) + (0.869 \times 0.5) + 0.1 \\ &= 0.82 \end{aligned}$$

④ Final Prediction

$$\text{output vector} = [1.30, 0.98, 0.82]$$

highest score $\Rightarrow 1.30 \Rightarrow$ predicted category:
classic

Output score

$$Z_c (\text{classic}) = 1.30$$

$$Z_{ch} (\text{chicken}) = 0.98$$

$$Z_v (\text{veggie}) = 0.82$$

Predicted category:

classic

+0.5